

Problem Set 2, COSC330/PHYS306
Sections 2-8, 2-11, and 3-2 through 3-6

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1 Chapter 2

1. [Section 2-8, Exercise 38] Convert each binary number to hexadecimal:

- (a) 1110
(b) 10
(c) 10111
(d) 10100110
(e) 1111110000
(f) 100110000010

2. [Section 2-8, Exercise 39] Convert each hexadecimal number to decimal:

- (a) 23_{16}
- (b) 92_{16}
- (c) $1A_{16}$
- (d) $8D_{16}$
- (e) $F3_{16}$
- (f) EB_{16}
- (g) $5C2_{16}$
- (h) 700_{16}

3. [Section 2-11, Exercise 56] Convert each binary number to Gray code:

- (a) 11011
(b) 1001010
(c) 1111011101110

4. [Section 2-11, Exercise 57] Convert each Gray code to binary:

- (a) 1010
(b) 00010
(c) 11000010001

2 Chapter 3

5. [Section 3-2, Exercise 6] Repeat Problem 5 for the waveforms in Figure 3-79.

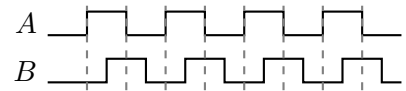


FIGURE 3-79

6. [Section 3-2, Exercise 7] The input waveforms applied to a 3-input AND gate are as indicated in Figure 3-80. Show the output waveform in proper relation to the inputs with a timing diagram.

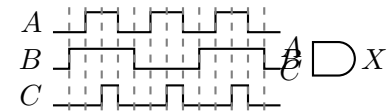


FIGURE 3-80

7. [Section 3-3, Exercise 12] Repeat Problem 8 for a 4-input OR gate.

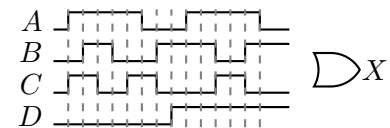


FIGURE 3-81

8. [Section 3-4, Exercise 17] For the set of input waveforms in Figure 3-83, determine the output for the gate shown and draw the timing diagram.



FIGURE 3-83

9. [Section 3-4, Exercise 18] Determine the gate output for the input waveforms in Figure 3-84 and draw the timing diagram.

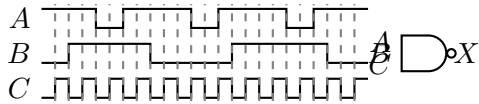


FIGURE 3-84

10. [Section 3-5, Exercise 22] Determine the output waveform in Figure 3-87 and draw the timing diagram.

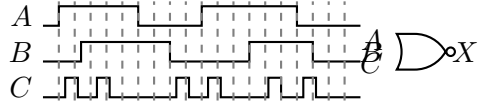


FIGURE 3-87

11. [Section 3-5, Exercise 24] The NAND and the negative-OR symbols represent equivalent operations, but they are functionally different. For the NOR symbol, look for at least one HIGH on the inputs to give a LOW on the output. For the negative-AND, look for two LOWs on the inputs to give a HIGH output. Using these two functional points of view, show that both gates in Figure 3-88 will produce the same output for the given inputs.

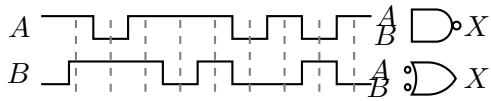


FIGURE 3-88

12. [Section 3-6, Exercise 26] Repeat Problem 17 for an exclusive-OR gate.
13. [Section 3-6, Exercise 27] Repeat Problem 17 for an exclusive-NOR gate.